

УТВЕРЖДАЮ: Коммерческий директор

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Technical specification for output shaft 45.67.89

This technical specification applies to the output shaft of gearbox R-160, drawing number 45.67.89, item code RD-450789 (hereinafter "the product"), manufactured by Rodos-Detal Group at its Kovrov site. The specification is used for export contract KMP-2025/02 with TOO «KazMashParts» (Republic of Kazakhstan) and for other export deliveries of the product, together with the Russian-language original TU-45-67-89.

1. Technical requirements

- 1.1. The product is made of steel grade 40X (GOST 4543-2016); blank — hot-rolled round bar, diameter 80 mm, length 420 mm.
- 1.2. The bearing journal diameter is machined to tolerance field $\phi 60\text{ k6}$.
- 1.3. Radial runout of the bearing journals shall not exceed 0.03 mm.
- 1.4. Surface roughness of the mating surfaces — $Ra\ 1.25\ \mu\text{m}$.
- 1.5. Hardness of the induction-hardened journals — 45...50 HRC.
- 1.6. Product weight — 11.4 kg.
- 1.7. No permanent coating is applied; for long-term storage the product is preserved with oil K-17 in accordance with quality-control method OTK-MI-008.
- 1.8. Marking includes the part designation, batch number and the quality-control stamp; placement and method are set out in the process route card.
- 1.9. Delivery set: the product in preservation packaging, together with a quality certificate.
- 1.10. The product surface shall be free of dents, cracks, corrosion and impact marks; isolated tooling marks that do not push the dimension outside the tolerance field are acceptable.
- 1.11. The product is manufactured at the Kovrov site along the route: turning, milling of the keyway, induction hardening, cylindrical grinding, quality-control inspection.

2. Safety requirements

- 2.1. The product presents no hazard during storage and transportation in standard packaging.
- 2.2. Installation and removal of the product within the gearbox shall follow the gearbox operating manual and the lifting rules applicable to loads over 10 kg.

3. Acceptance rules

- 3.1. The manufacturer's quality-control department accepts the product by batch on the basis of the accompanying documentation.
- 3.2. The buyer's incoming inspection by quantity and quality is performed under the acceptance procedure set out in the supply contract.
- 3.3. Claims are submitted to the supplier under the terms of contract KMP-2025/02, stating the batch number.
- 3.4. Sample inspection of dimensions at batch acceptance by the manufacturer covers not less than 5 % of the batch, and not less than three units; if a nonconformity is found in any unit, the sample size is doubled.

4. Control methods

- 4.1. Linear dimensions and tolerance fields are controlled under method OTK-MI-001.
- 4.2. Radial runout and shaft geometry are controlled under method OTK-MI-002.
- 4.3. Hardness after heat treatment is controlled under method OTK-MI-003.
- 4.4. Surface roughness is controlled under method OTK-MI-004.

5. Transportation and storage

- 5.1. The product may be transported by any covered mode of transport that prevents mechanical damage and moisture ingress.
- 5.2. For export deliveries under contract KMP-2025/02 the product is delivered DAP Almaty.
- 5.3. The product is stored in a closed, heated warehouse at a temperature not below 5 °C and relative humidity not exceeding 70 %.
- 5.4. Storage life without re-preservation does not exceed 12 months from the shipment date.

6. Operating instructions

- 6.1. Installation into the R-160 gearbox is performed in accordance with the gearbox operating manual.
- 6.2. Mating surfaces are cleaned of preservation coating and inspected for damage before installation.
- 6.3. Operation of the product outside the rated conditions of the R-160 gearbox is not permitted.
- 6.4. If the bearing units are removed during gearbox maintenance, it is recommended to re-check the radial runout of the shaft journals before reassembly.

7. Manufacturer's warranty

- 7.1. The warranty period for the product is 12 months from the shipment date.
- 7.2. The warranty does not cover failures caused by improper installation, operation outside rated conditions, or mechanical damage.
- 7.3. Warranty obligations terminate if the product is used together with equipment that does not conform to the R-160 gearbox operating documentation.